INTERNSHIP COMPLETION REPORT

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Internship Duration : 8 Weeks

Intern Details

Batch Number : 2

Start Date: JULY 2026

Internship Duration: 8 Weeks

Project Domain: Data Visualization

Tools Used: Power BI, Python (Pandas)

1. Project Title

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

2. Project Objective

The primary objective of this project is to develop an AI-Powered Workforce Analytics & Talent Intelligence Dashboard that enables organizations to transform workforce-related data into actionable business intelligence. The platform is designed to provide comprehensive insights into workforce performance, employee engagement, talent development, diversity, attrition, recruitment effectiveness, and overall organizational health.

The project aims to:

- Integrate workforce data from different HR and talent-related sources.
- Clean, transform, and model workforce data for analytical use.
- Analyze workforce performance, demographics, engagement, attrition, and recruitment trends.
- Develop predictive analytics models for identifying employee attrition risks.
- Identify skill gaps and support employee career development.
- Generate workforce health scores and talent risk indicators.

- Provide AI-powered recommendations for workforce planning and employee engagement.
- Enable natural-language interaction with workforce data.
- Use LLM, RAG, predictive analytics, and agentic AI approaches to provide intelligent workforce insights.
- Develop an interactive dashboard for HR leaders and business stakeholders.

3. Project Description in Detail

3.1 Overview

The AI-Powered Workforce Analytics & Talent Intelligence Dashboard is designed as an intelligent workforce analytics platform that combines data engineering, analytics, machine learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI workflows.

Organizations generate workforce information from multiple systems, including HR management systems, recruitment platforms, learning management systems, performance reviews, and employee engagement surveys. Analyzing these sources independently can make it difficult for HR leaders to obtain a complete view of organizational health.

The proposed platform brings workforce information together and transforms it into meaningful insights through dashboards, predictive analytics, AI-generated recommendations, and conversational querying.

The system focuses on workforce performance, employee engagement, talent development, diversity, attrition, recruitment effectiveness, skill gaps, and workforce planning.

3.2 Data Collection & Preparation

The proposed system can work with workforce information obtained from sources such as:

- SAP SuccessFactors
- Workday
- Oracle HCM
- Recruitment portals
- Learning Management Systems
- Employee engagement surveys
- Performance review systems

The data may contain information related to:

- Employee demographics
- Department/domain
- Salary

- Experience
- Performance
- Skills
- Learning progress
- Job satisfaction
- Work-life balance
- Attrition
- Recruitment
- Engagement
- Diversity

Data Preparation

The data preparation process includes:

- Data collection
- Data integration
- Data cleansing
- Missing-value handling
- Duplicate removal
- Data transformation
- Data validation
- Metadata extraction
- Feature engineering
- Workforce data modeling

The final report should include your actual dataset name, number of records, number of columns, data sources, and preprocessing results here.

3.3 Data Cleaning (Python Pandas)

The workforce data is organized into a structured model to support efficient analytics and reporting.

The data model can include:

- Employee information
- Department information
- Performance information

- Learning information
- Skills information
- Attrition information
- Recruitment information
- Engagement information

A suitable architecture can be represented as:

Data Sources → Data Integration → Data Cleaning → Data Modeling → Workforce Intelligence Repository → Analytics & AI → Dashboard

3.3 Website Development

The Workforce Data Model was developed as an important module of the website to organize and manage workforce-related information in a structured manner. The website provides a centralized interface where different types of employee and organizational data can be stored, accessed, and analyzed efficiently. The module includes employee information, department details, performance data, learning and training information, employee skills, attrition details, recruitment information, and employee engagement data. These data categories are integrated into the web application to provide a complete view of the workforce and support effective analytics and reporting.

The website interface is designed to make workforce data easy to view and manage. Employee information includes details such as employee ID, age, job role, department, and other relevant attributes, while department information helps organize employees according to their respective departments and roles. Performance and learning information are used to monitor employee performance, training activities, certifications, and development progress. The skills section helps identify employee expertise and skill gaps, while attrition information supports the analysis of employee turnover. Recruitment and engagement information provide insights into hiring activities and employee satisfaction. By integrating these data categories, the website supports dashboards, analytics, AI-based recommendations, and decision-support features, enabling management to obtain meaningful workforce insights and make better data-driven decisions.

The website consists of the following major components:

1. Home Page

The Home Page provides an overview of the AI-Powered Workforce Analytics and Talent Intelligence Dashboard. It gives users quick access to the main modules, analytics, AI assistant, recommendations, and workforce insights.

2. AI Assistant – RAG

The AI Assistant uses Retrieval-Augmented Generation (RAG) to answer user queries based on uploaded workforce documents and datasets. It retrieves relevant information and provides contextual, data-driven responses.

3. Admin Decision Assistant

The Admin Decision Assistant helps administrators make informed workforce decisions by analyzing employee data, identifying important trends, and providing AI-based insights and recommendations.

4. AI Recommendations Page

The AI Recommendations page presents personalized recommendations and actionable insights generated from workforce data. It helps management identify areas for improvement and take suitable actions.

5. Data Management Page

The Data Management page allows administrators to upload, manage, view, and organize workforce datasets and documents used for analytics and AI-based processing.

6. Executive Dashboard

The Executive Dashboard provides a high-level summary of workforce performance using key metrics, charts, and visualizations. It helps executives quickly understand organizational trends and make strategic decisions.

7. Workforce & Demographic Data

This page displays workforce information based on employee demographics and organizational characteristics, such as age, gender, department, experience, and other workforce attributes.

8. Salary & Experience Insights

The Salary & Experience Insights page analyzes the relationship between employee salary, experience, roles, and other workforce factors. Visualizations help identify salary patterns and experience-based trends.

9. Attrition & Employee Insights

The Attrition & Employee Insights page provides analysis of employee attrition patterns and workforce factors. It helps identify potential attrition trends and provides insights that can support employee retention strategies.

3.4 Technology Stack

The technology stack used for the project consists of tools and technologies for data processing, analytics, machine learning, artificial intelligence, dashboard development, and web application development.

- **Python:** Used for workforce data processing, data analysis, preprocessing, and machine learning implementation.
- **Pandas:** Used for data cleaning, transformation, manipulation, and preparation of workforce datasets.
- **NumPy:** Used for numerical operations and data processing.
- **Scikit-learn:** Used for implementing and evaluating machine learning models, where applicable.
- **[Dashboard Technology]:** Used for developing interactive workforce dashboards, KPI cards, charts, filters, and workforce visualizations.
- **[Frontend Technology]:** Used to develop the web-based user interface and interactive components.
- **[Backend Technology]:** Used for implementing APIs, business logic, data communication, and integration between the frontend and analytical components.
- **[Database]:** Used for storing and managing workforce-related information.

- **Large Language Model (LLM):** Used to support natural-language interaction and intelligent workforce insights.
- **Retrieval-Augmented Generation (RAG):** Used, where implemented, to retrieve relevant workforce information and provide contextual information to the language model.
- **[AI/Agent Framework]:** Used, where implemented, for developing agentic AI workflows and specialized workforce intelligence agents.
- **[Cloud/Deployment Platform]:** Used for application deployment, if applicable.

Data Sources

The platform is designed to work with workforce data from sources such as:

- SAP SuccessFactors
- Workday
- Oracle HCM
- Recruitment platforms
- Learning Management Systems
- Employee engagement surveys
- Performance review systems
- Project-provided workforce datasets

3.5 Real-World Impact

The The AI-Powered Workforce Analytics & Talent Intelligence Dashboard can help organizations transform workforce data into actionable insights for strategic decision-making. By combining workforce analytics, predictive analytics, and AI-powered intelligence, the platform supports HR teams in identifying workforce challenges and planning appropriate interventions.

1. Workforce Planning

The platform provides workforce trends, headcount analysis, and predictive insights that can support organizations in planning future workforce requirements and resource allocation.

2. Attrition and Retention Management

Attrition analytics and predictive risk indicators can help HR teams identify workforce groups that may have a higher risk of employee turnover and develop proactive retention strategies.

3. Talent Development

Skill-gap analysis provides visibility into existing workforce capabilities and potential skill shortages. These insights can support training, reskilling, upskilling, and career-development initiatives.

4. Employee Engagement

The platform analyzes available employee engagement, satisfaction, and work-life-balance indicators to help organizations identify areas requiring improvement.

5. Diversity Monitoring

The dashboard provides visibility into available workforce demographic and diversity metrics, helping stakeholders monitor organizational diversity patterns and identify areas requiring attention.

6. Recruitment Effectiveness

Recruitment analytics can help organizations understand recruitment funnel performance, hiring trends, and workforce acquisition effectiveness.

7. AI-Driven Decision Support

AI-powered recommendations can transform workforce analytics into actionable suggestions related to retention, learning, engagement, talent development, and workforce planning.

8. Conversational Workforce Intelligence

Natural-language interaction allows HR leaders and stakeholders to ask questions about workforce data without manually analyzing multiple dashboards or reports.

9. Organizational Health Monitoring

Workforce health scores and talent-risk indicators can provide stakeholders with a consolidated view of organizational workforce conditions and help identify potential risks at an early stage.

Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understanding project scope, workforce data requirements, and analytics specifications	Studied the project statement, identified workforce analytics requirements, understood expected outcomes, and finalized the initial project scope.
Week 2	Dataset collection, data examination, and preprocessing	Collected/selected the available workforce dataset, examined the attributes, performed data cleaning, handled missing/inconsistent data, and prepared the dataset for analysis.
Week 3	Workforce data modeling and exploratory analysis	Developed the initial workforce data model and performed exploratory analysis to identify workforce, performance, engagement, and attrition patterns.
Week 4	Predictive analytics and AI engine development	Developed and evaluated the applicable analytics/predictive components for attrition analysis, workforce trends, and skill-gap identification.
Week 5	Workforce intelligence dashboard development	Developed dashboard components for workforce overview, headcount, attrition, engagement, diversity, and other available workforce metrics.
Week 6	Dashboard enhancement and AI integration	Integrated analytical results into the dashboard and developed applicable AI-powered insights, recommendations, or conversational functionality.

Week 7	Testing, validation, and system refinement	Conducted functional testing, data validation, model evaluation, dashboard testing, and refinement of identified issues.
Week 8	Documentation, final reporting, and presentation	Completed project documentation, prepared the final report and presentation, and prepared the project for final demonstration.

4a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project requirements studied, workforce analytics scope defined, and system objectives identified.	Week 1
Data Foundation	Workforce dataset selected, cleaned, transformed, and prepared for analytics.	Week 2
Data Modeling Complete	Workforce data model and analytical structure developed.	Week 3
AI Analytics Engine	Workforce analytics and applicable predictive models developed and evaluated.	Week 4
Dashboard Development	Workforce intelligence dashboard developed with key workforce visualizations and KPIs.	Week 5
AI Integration	AI-powered recommendations and/or conversational workforce intelligence integrated, where implemented.	Week 6
Testing & Validation	System, dashboard, data, and analytical components tested and refined.	Week 7
Final Submission	Final documentation, presentation, demonstration, and project deliverables completed.	Week 8

4b. Project Execution Details

Phase 1: Data Acquisition & Preparation

- **Requirement Analysis:** Studied the workforce analytics requirements and identified the major areas of analysis, including workforce performance, attrition, engagement, talent development, diversity, recruitment, and skill gaps.
- **Data Collection:** Identified and collected the available workforce dataset required for developing the analytics solution.
- **Data Cleaning:** Examined the dataset for missing values, duplicate records, inconsistent values, incorrect data types, and other data-quality issues.

- **Data Transformation:** Standardized and transformed the workforce data into a structured format suitable for analytics and visualization.
- **Feature Preparation:** Prepared relevant workforce attributes and analytical features required for descriptive and predictive analysis.

Phase 2: Workforce Analytics & AI Development

- **Workforce Analytics:** Developed analytical components to understand workforce demographics, performance, engagement, headcount, attrition, recruitment, and other available workforce indicators.
- **Attrition Analysis:** Analyzed historical workforce patterns to identify factors and groups associated with employee attrition.
- **Predictive Analytics:** Developed and evaluated applicable machine learning models for predicting attrition risk and supporting proactive workforce decisions.
- **Skill Gap Analysis:** Analyzed available employee skills and organizational requirements to identify potential skill gaps and learning needs.
- **Workforce Health:** Developed workforce health indicators based on relevant workforce metrics.
- **Talent Recommendations:** Generated workforce recommendations based on analytical results and available AI capabilities.

Phase 3: Dashboard Development

- **Workforce Overview:** Developed an interactive overview containing important workforce KPIs and organizational statistics.
- **Headcount Analysis:** Created visualizations to understand workforce distribution and headcount trends.
- **Attrition Dashboard:** Developed visualizations for attrition trends, rates, and available risk indicators.
- **Engagement Dashboard:** Developed visualizations for employee engagement, satisfaction, and related workforce metrics.
- **Diversity Dashboard:** Created visualizations for available workforce diversity and demographic metrics.
- **Recruitment Dashboard:** Developed available recruitment funnel and hiring analytics.
- **Skill Gap Dashboard:** Created visualizations to highlight workforce skill gaps and development requirements.

Phase 4: AI Integration & Website Development

- **Front-End Implementation:** Developed the web-based interface for accessing workforce analytics and talent intelligence modules using **[actual frontend technologies]**.
- **Dashboard Integration:** Integrated workforce analytics and visualization components into the web application.
- **AI Integration:** Integrated the implemented AI capabilities for generating workforce insights and recommendations.
- **Conversational Intelligence:** Developed/integrated a natural-language interface, where implemented, to allow users to query workforce information.
- **RAG Integration:** Implemented the RAG workflow, where applicable, to retrieve relevant workforce information and provide contextual AI responses.
- **Report Generation:** Developed reporting functionality for presenting workforce analytics and strategic insights, where implemented.

Phase 5: Testing & Final Delivery

- **System Testing:** Tested the functionality of the major website and dashboard components.
- **Data Validation:** Verified that the displayed analytics and visualizations corresponded to the underlying workforce data.
- **Model Evaluation:** Evaluated applicable predictive models using appropriate performance metrics.
- **AI Validation:** Tested AI-generated responses and recommendations for relevance and consistency.
- **Bug Resolution:** Identified and resolved implementation issues discovered during testing.
- **Final Documentation:** Prepared the technical documentation, project report, screenshots, and final presentation.
- **Final Demonstration:** Prepared the completed solution for the Infosys Virtual Internship 7.0 final demonstration.

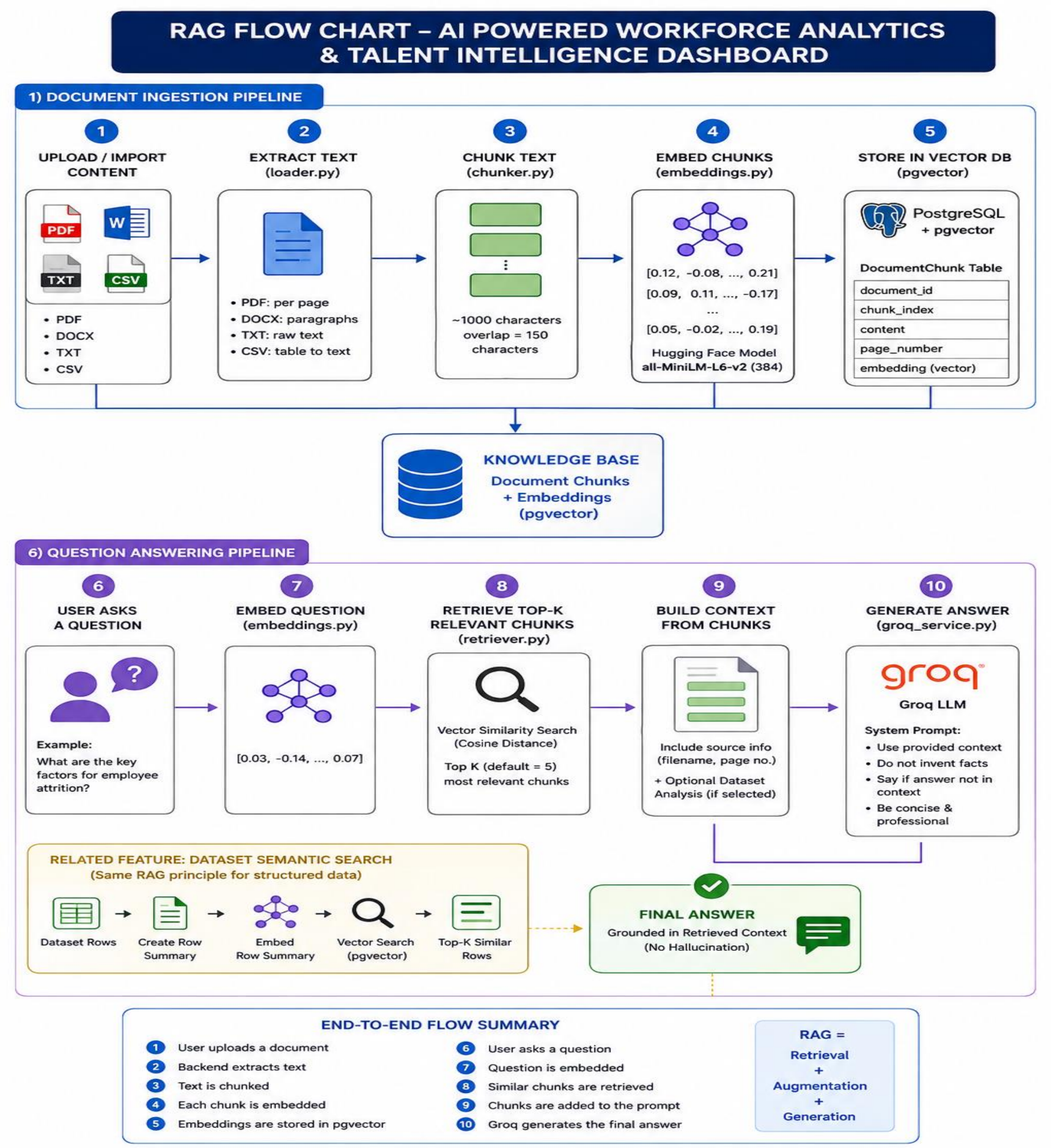

Figure 5.1: RAG Workflow for AI-Powered Workforce Analytics & Talent Intelligence Dashboard

5. Snapshots / Screenshots

Dashboard/Page	Screenshot Reference	Description
Home Page	Fig 1.1	The landing page of the website, setting the theme of global inequality.
Country Data Cards	Fig 1.2	An interactive component displaying key inequality data for different countries at a glance.
Inequality Report	Fig 1.3	The report generation tool for India in 2022, showing a Gini of 33 and an AI-generated insight.
Login / Register	Fig 1.4	The user authentication page for accessing full platform features.
About	Fig 1.5	The "About" page, detailing the project's mission, data sources, and target audience.
Main Dashboard	Fig 1.6	The core "Global Income Inequality Dashboard" with KPIs, trend lines, and global distribution charts.
Regional Analysis	Fig 1.7	The regional analysis page, highlighting comparisons, trends, and GDP correlation.
Country Comparison	Fig 1.8	The detailed country view, allowing for deep dives into specific nations with trend lines and income distribution charts.

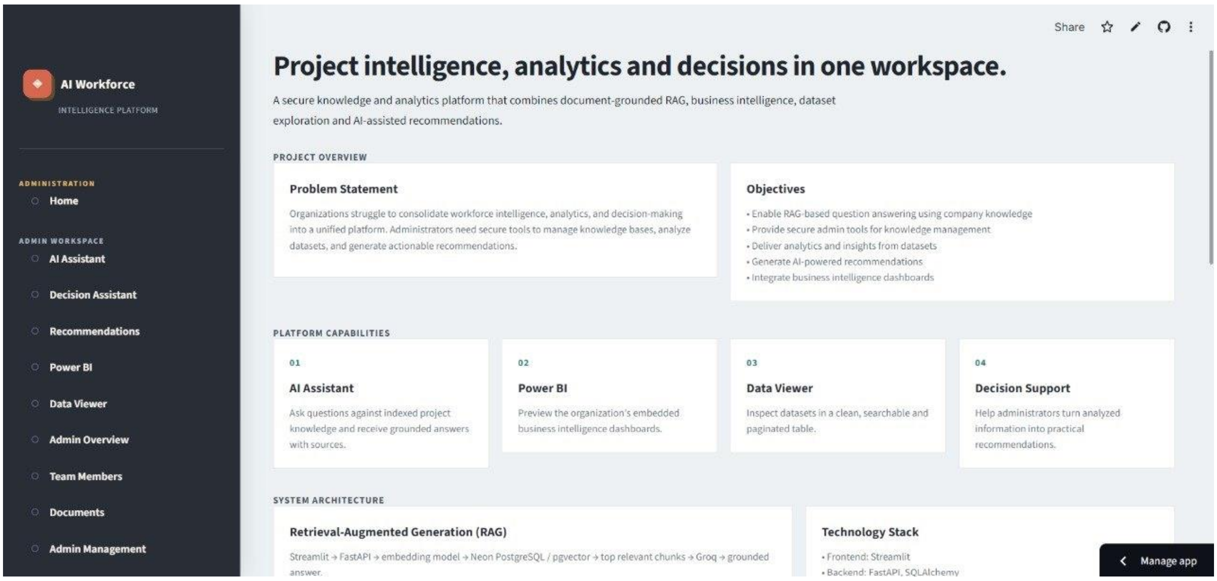

Fig 1.1 : Home Page

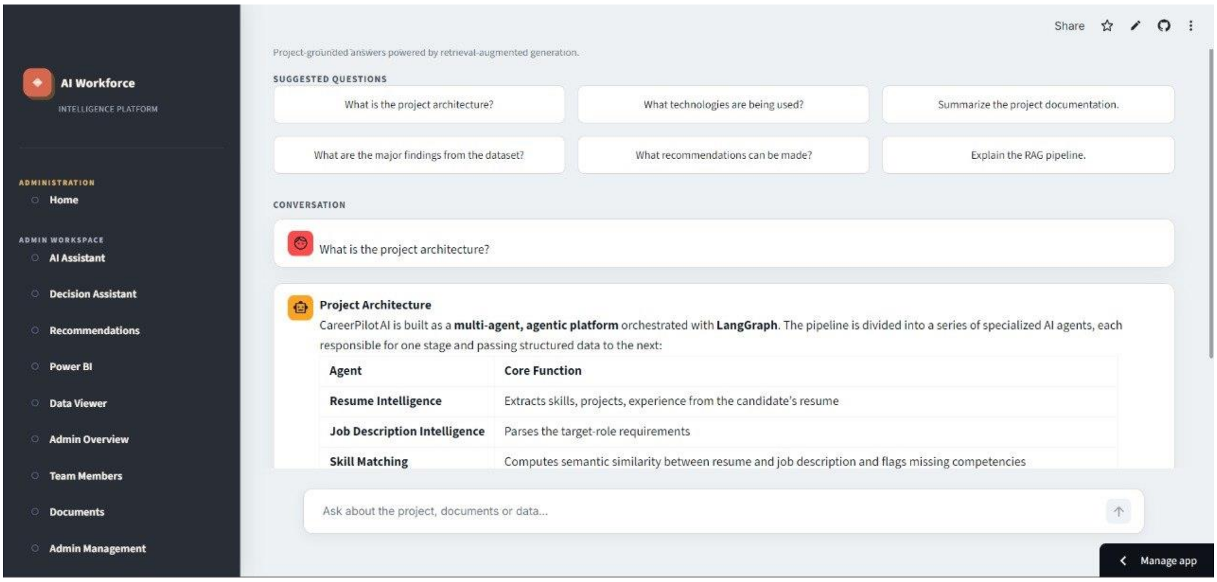

Fig 1.2 : AI Assistant RAG

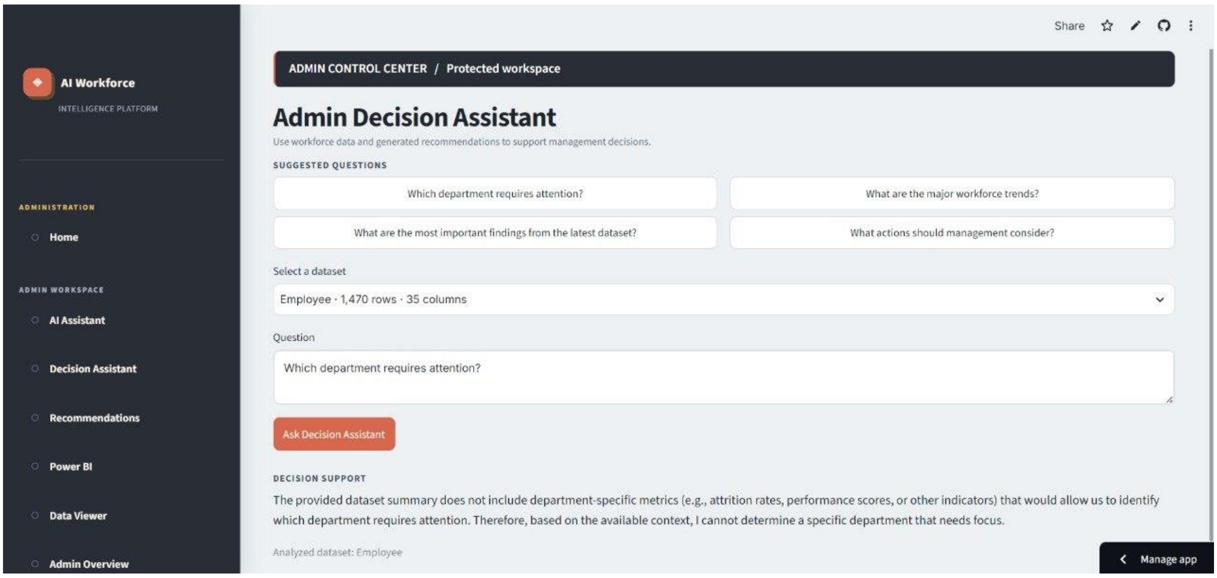

Fig 1.3 : Admin Decision Assistant

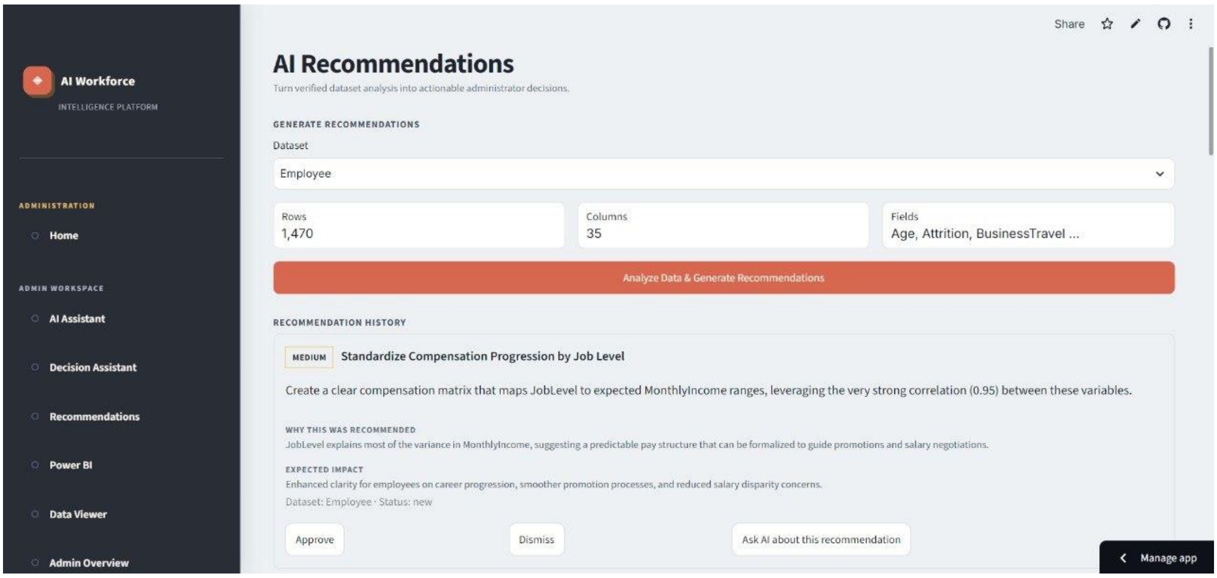

Fig 1.4 : AI Recommendations page

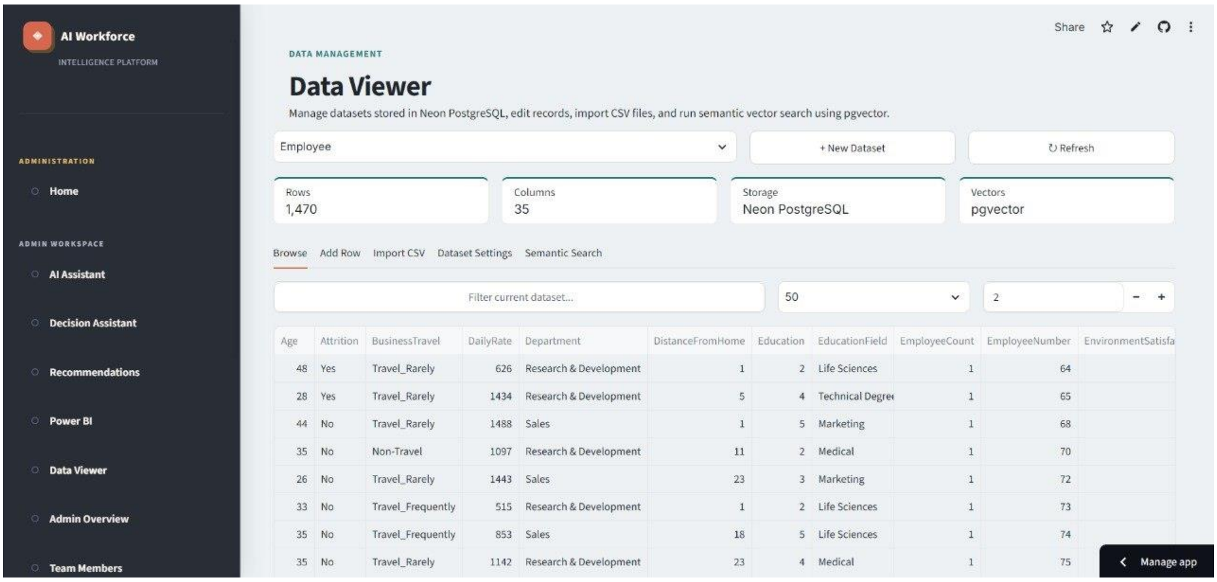

Fig 1.5 : Data management page

PowerBI Dashboards :

Fig 1.6 : Executive Dashboard

Fig 1.7 : Workforce & Demographic Data

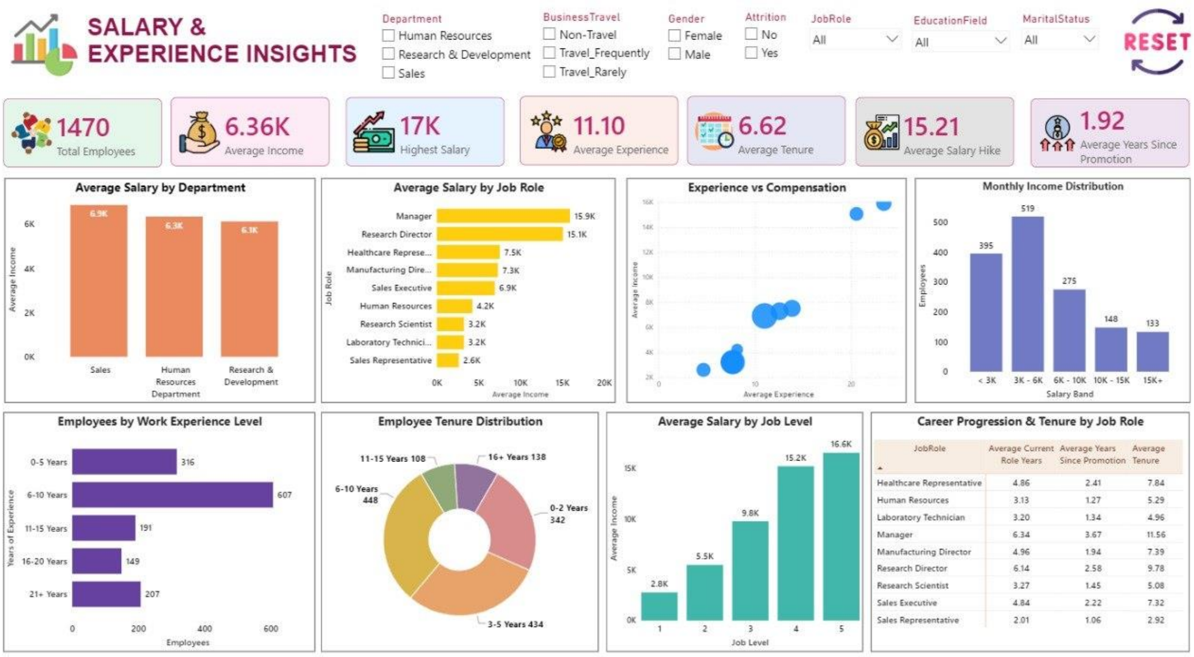

Fig 1.8 : Salary & Experience Insights

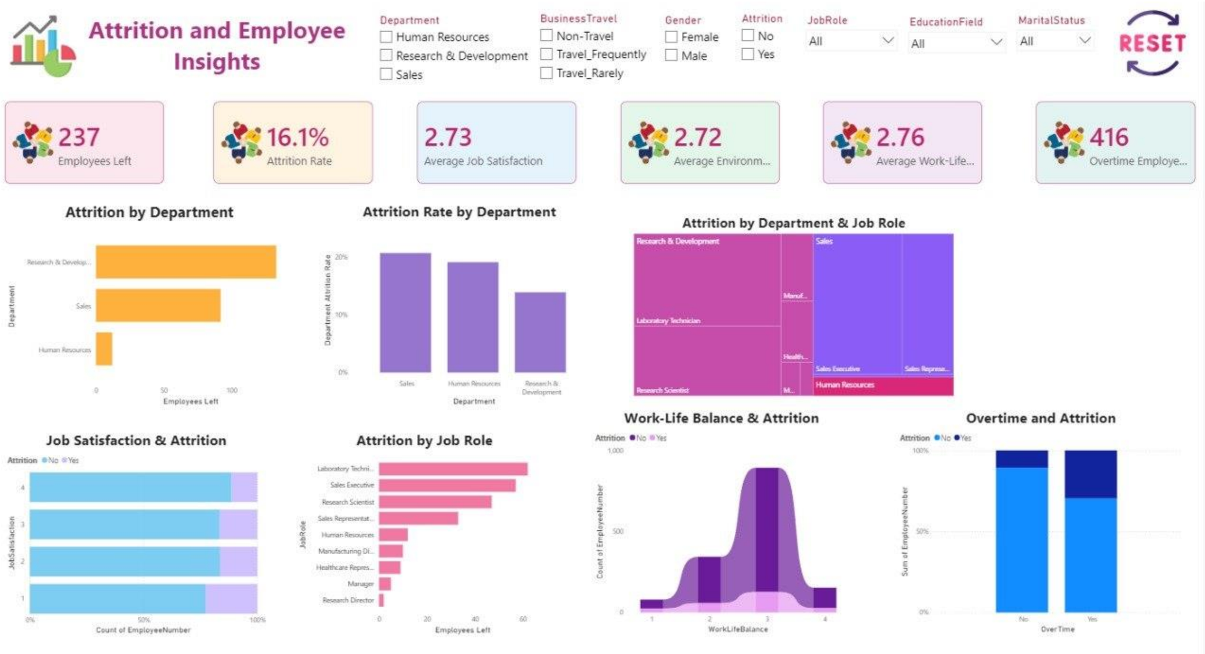

Fig 1.9 : Attrition & Employee Insights

6. Challenges Faced

Data Inconsistency: Workforce data can originate from different HR systems, recruitment platforms, learning systems, and engagement surveys, resulting in differences in formats, attributes, missing values, and data definitions. **Resolution:** We performed data cleaning, standardization, transformation, and validation to prepare the available workforce datasets for consistent analysis. Common data formats and appropriate preprocessing techniques were applied before integrating the data into the analytics workflow.

Missing and Incomplete Workforce Data: Some workforce attributes may contain missing or incomplete values, which can affect analytical and predictive results. **Resolution:** We identified missing values during the preprocessing stage and applied appropriate data-cleaning and validation techniques based on the characteristics of the available dataset.

Predictive Analytics Complexity: Predicting employee attrition and identifying workforce risks can be challenging because employee behavior is influenced by multiple factors such as performance, satisfaction, experience, workload, and engagement. **Resolution:** We analyzed relevant workforce attributes and evaluated the implemented predictive models using appropriate validation techniques to improve the reliability of the analytical results.

AI Response Reliability: Integrating AI capabilities into workforce analytics requires relevant and reliable contextual information. Incorrect or incomplete context can result in less useful AI-generated responses.

Resolution: The project uses the implemented retrieval and contextual-processing approach to provide relevant workforce information to the AI component before generating responses and recommendations.

Dashboard Complexity: Workforce analytics involves multiple dimensions such as attrition, engagement, performance, skills, recruitment, diversity, and workforce planning. Presenting all these metrics without making the dashboard difficult to understand was a challenge.

Resolution: We organized the platform into separate analytical modules and used KPI cards, charts, filters, and focused dashboard sections to make the information easier to understand.

Integration of Multiple Project Components: Integrating data processing, analytics, machine learning, dashboard visualization, and AI capabilities into a single platform required careful coordination between different components.

Resolution: We followed a modular development approach and tested individual components before integrating them into the overall workforce intelligence platform.

7. Learnings & Skills Acquired

Tools & Technologies: Gained practical experience in workforce data processing, analytics, machine learning, dashboard development, and web-based application development. We also gained exposure to AI technologies such as Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI-powered recommendation workflows.

Data Analytics: Learned how to collect, clean, transform, validate, and analyze workforce data. We gained practical understanding of workforce indicators such as headcount, attrition, employee engagement, performance, skills, recruitment, and organizational trends.

Artificial Intelligence & Machine Learning: Developed an understanding of applying predictive analytics to workforce-related problems such as attrition prediction and skill-gap identification. We also learned how AI can be used to transform analytical results into meaningful recommendations.

Dashboard Development: Learned how to convert analytical results into interactive visualizations and dashboards. We gained experience in presenting workforce KPIs, trends, comparisons, risk indicators, and recommendations in a user-friendly format.

LLM, RAG & Conversational Analytics: Gained an understanding of how Large Language Models and Retrieval-Augmented Generation can be incorporated into workforce intelligence systems to enable natural-language interaction and context-aware responses.

Problem Solving: Improved our ability to identify technical and data-related challenges and develop appropriate solutions during different stages of project implementation.

Teamwork & Communication: Enhanced collaboration and communication skills through team discussions, task allocation, knowledge sharing, progress updates, and collaborative problem-solving throughout the internship.

Domain Knowledge: Developed practical knowledge of workforce analytics and talent intelligence, including employee attrition, engagement, talent development, skill gaps, workforce planning, recruitment effectiveness, diversity metrics, and organizational health.

8.Testimonials from Team

This internship provided me with valuable hands-on experience in transforming workforce data into meaningful insights. Working on the AI-Powered Workforce Analytics & Talent Intelligence Dashboard helped me understand how data analytics, machine learning, and artificial intelligence can be applied to real-world HR and workforce challenges. I gained practical experience in data preprocessing, dashboard development, and analytical problem-solving. Working as part of a team also improved my communication, collaboration, and project-management skills. The experience gave me greater confidence in applying my technical knowledge to real-world business problems.

9.Conclusion

The Infosys Virtual Internship 7.0 provided us with a valuable opportunity to develop the AI-Powered Workforce Analytics & Talent Intelligence Dashboard, a solution designed to transform workforce data into actionable organizational intelligence.

Throughout the project, we worked on different stages of the solution, including workforce data preparation, data analysis, predictive analytics, dashboard development, AI integration, testing, and documentation. The platform brings together important workforce dimensions such as employee performance, headcount, attrition, engagement, diversity, recruitment, skills, talent risks, and workforce health.

The integration of AI capabilities provides an additional layer of intelligence by enabling data-driven recommendations and natural-language interaction with workforce information, where implemented. These capabilities can help HR leaders and business stakeholders understand workforce trends, identify potential risks, recognize skill gaps, and support better workforce planning and talent-development decisions.

The project also provided us with practical experience in data analytics, machine learning, artificial intelligence, dashboard development, teamwork, communication, problem-solving, and technical documentation.

Overall, the internship was a valuable learning experience that helped us bridge the gap between academic knowledge and real-world project development. The project has strengthened our technical and professional skills and provided us with practical exposure to applying AI and analytics to workforce and business challenges.

10.Acknowledgements

We would like to express our sincere gratitude to Infosys and Infosys Springboard for providing us with the opportunity to participate in Infosys Virtual Internship 7.0 and work on the AI-Powered Workforce Analytics & Talent Intelligence Dashboard.

We are deeply grateful to our mentors, internship coordinators, and trainers for their valuable guidance, technical support, feedback, and encouragement throughout the project. Their guidance helped us understand the project requirements and improve our technical approach at different stages of development.

We would also like to thank all our team members for their hard work, collaboration, knowledge sharing, communication, and dedication throughout the internship. Each member's contribution played an important role in progressing the project towards completion.

Finally, we would like to thank everyone who supported and encouraged us during the internship. This experience has provided us with valuable technical knowledge, professional skills, and practical exposure that will contribute to our future careers.

Thank you to everyone who contributed to the successful completion of this project.